

We are given the cubic function  $y = x^3 + 2x^2 + 4x + 2$ , and we want to know its roots. We can solve this question in a few ways, here are two.

For both methods we notice the equation has a nonzero quadratic term, so we depress it by making the translation  $x \rightarrow x - \frac{2}{3}$ . This gives us

$$\begin{aligned} & \left(x - \frac{2}{3}\right)^3 + 2\left(x - \frac{2}{3}\right)^2 + 4\left(x - \frac{2}{3}\right) + 2 \\ &= x^3 - 2x^2 + \frac{4}{3}x - \frac{8}{27} + 2x^2 - \frac{8}{3}x + \frac{8}{9} + 4x - \frac{8}{3} + 2 \\ &= x^3 + \frac{8}{3}x - \frac{2}{27}. \end{aligned}$$

## 1 First Method

The historical development of the solutions to cubic equations is intimately tied to the complex numbers. One in particular that we will require is  $e^{\frac{2\pi i}{3}} = \frac{-1+i\sqrt{3}}{2} \equiv \omega$  which is a root of the equation  $x^3 = 1$ . Discarding the trivial root  $x = 1$  we get the useful characteristic polynomial of  $\omega$

$$1 + \omega + \omega^2 = 0.$$

We can use this complex number to state the following identity

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x + \omega y + \omega^2 z)(x + \omega^2 y + \omega z)$$

Since the left hand side is symmetric in the variables, so is the right hand side.

$$x^3 + y^3 + z^3 - 3xyz = (y + z + x)(y + \omega z + \omega^2 x)(y + \omega^2 z + \omega x)$$

$$x^3 + y^3 + z^3 - 3xyz = (z + x + y)(z + \omega x + \omega^2 y)(z + \omega^2 x + \omega y)$$

If we look at this as a polynomial in  $x$  and set it equal to our depressed cubic

$$x^3 + (-3yz)x + y^3 + z^3 = x^3 + \frac{8}{3}x - \frac{2}{27}$$

then we can conclude the roots are

$$-y - z, \quad -\omega y - \omega^2 z, \quad -\omega^2 y - \omega z$$

Using the above identity. Comparing coefficients we get the system of equations  $\frac{8}{3} = -3yz$  and  $-\frac{2}{27} = y^3 + z^3$ . We want to use the fact that  $(t - a)(t - b) = t^2 - (a + b)t + ab$ . Taking our first equation, dividing it by  $-3$ , and cubing both sides we now have

$$-\frac{2}{27} = y^3 + z^3 \quad -\frac{8^3}{9^3} = y^3 z^3$$

which lets us solve for  $y^3$  and  $z^3$  with the equation

$$t^2 + \frac{2}{27}t - \frac{8^3}{9^3}$$

$$\Rightarrow t = \frac{\frac{-2}{27} \pm \sqrt{\frac{2^2}{27^2} + 4 \cdot \frac{8^3}{9^3}}}{2} = \frac{-1 \pm 3\sqrt{57}}{27}$$

$$\Rightarrow y = \frac{\sqrt[3]{-1 + 3\sqrt{57}}}{3}, \quad z = \frac{\sqrt[3]{-1 - 3\sqrt{57}}}{3}.$$

This gives us that the roots of  $x^3 + \frac{8}{3}x - \frac{2}{27}$  are

$$\frac{-\sqrt[3]{-1 + 3\sqrt{57}} - \sqrt[3]{-1 - 3\sqrt{57}}}{3} \approx 0.02777,$$

$$\frac{-\omega \sqrt[3]{-1 + 3\sqrt{57}} - \omega^2 \sqrt[3]{-1 - 3\sqrt{57}}}{3} \approx -0.0139 - 1.6332i,$$

$$\frac{-\omega^2 \sqrt[3]{-1 + 3\sqrt{57}} - \omega \sqrt[3]{-1 - 3\sqrt{57}}}{3} \approx -0.0139 + 1.6332i.$$

If we invert our translation by sending  $x \rightarrow x + \frac{2}{3}$  then we get that the roots of the original equation  $x^3 + 2x^2 + 4x + 1$  are

$$\frac{2 - \sqrt[3]{-1 + 3\sqrt{57}} - \sqrt[3]{-1 - 3\sqrt{57}}}{3},$$

$$\frac{2 - \omega \sqrt[3]{-1 + 3\sqrt{57}} - \omega^2 \sqrt[3]{-1 - 3\sqrt{57}}}{3},$$

$$\frac{2 - \omega^2 \sqrt[3]{-1 + 3\sqrt{57}} - \omega \sqrt[3]{-1 - 3\sqrt{57}}}{3}$$

## 2 A crash course in hyperbolic complex numbers

Another way of generating the solutions to a cubic equation involves expanding our view of what a complex numbers are.

One way of thinking about  $\mathbb{C}$  from algebra is that it is isomorphic to  $\mathbb{R}[x]/x^2 + 1$ . Since  $x^2 + 1$  is irreducible over  $\mathbb{R}[x]$  we have that  $\mathbb{C}$  is a field.

If we modify this construction slightly to consider  $\mathbb{R}[x]/x^2 - 1$  we get something isomorphic to  $\mathbb{R}[j]$  where  $j^2 = 1$  and  $j \neq \pm 1$ . We will call this object  $\mathbb{H}$ , the complex hyperbolic plane. Since  $x^2 - 1$  is not irreducible over  $\mathbb{R}[x]$  we have  $\mathbb{H}$  is not a field, but if an element  $a + bj \in \mathbb{H}$

is not of the form  $a \pm aj$ , for  $a \in \mathbb{R}$ , then  $(a + bj)^{-1} = \frac{a-bj}{a^2-b^2}$ . Lots of theorems of the complex numbers have a hyperbolic sibling. For instance, we have an analog of Euler's formula

$$e^{jt} = \cosh t + j \sinh t.$$

There are two key differences between  $\mathbb{H}$  and  $\mathbb{C}$ . The first is that a polynomial of degree  $n$  can have at most  $n^2$  distinct roots over  $\mathbb{H}$ , for example  $x^2 - 4$  has roots  $\pm 2, \pm 2j$ . The second is that the hyperbolic complex numbers are spanned by the basis  $u_+ = \frac{1}{2}(1 + j)$  and  $u_- = \frac{1}{2}(1 - j)$  which has the powerful properties  $u_+^2 = u_+$ ,  $u_-^2 = u_-$  (idempotent) and  $u_+u_- = 0$  (mutually annihilating). These numbers can be identified as direction vectors of the isotropic null lines and so  $\{u_+, u_-\}$  is called the null basis.

To use the special computational properties of this basis, we also will need some notation. For a number  $z = x + yj$ , we define

$$z_+ = x + y \quad z_- = x - y$$

then we have

$$z = z_+u_+ + z_-u_-$$

as well as

$$zu_+ = z_+u_+ \quad zu_- = z_-u_-.$$

### 3 Second Method

The following treatment is taken from "New foundations in mathematics: the geometric concept of number" by Garret Sobczyk.

If we allow  $z = a + bj$  to have complex coefficients  $a, b \in \mathbb{C}$  then Euler's formula changes slightly to  $e^{ijt} = \cos t + ji \sin t$  and we get the following theorem

**Theorem 1.** *The basic unipodal equation  $z^n = r$  has the  $n^2$  solutions*

$$z = (e^{\frac{i2\pi}{n}})^k r_+^{\frac{1}{n}} u_+ + (e^{\frac{i2\pi}{n}})^l r_-^{\frac{1}{n}} u_-$$

where  $k, l = 0, 1, 2, \dots, n-1$  and  $r_{\pm}^{\frac{1}{n}}$  is any  $n^{\text{th}}$  root of  $r_{\pm}$ .

Proof Using the idempotent property of the null basis we can say that

$$\begin{aligned} z^n &= z_+^n u_+ + z_-^n u_- = r_+ + r_- \\ \implies z_+^n &= r_+, \quad z_-^n = r_- \end{aligned}$$

which are two equations over  $\mathbb{R}$ . Thus they both have the solutions

$$z_+ = (e^{\frac{i2\pi}{n}})^k r_+^{\frac{1}{n}}, \quad z_- = (e^{\frac{i2\pi}{n}})^l r_-^{\frac{1}{n}}$$

for  $k, l = 0, 1, 2, \dots, n-1$ . Since  $z = z_+ u_+ + z_- u_-$  we achieve the result.

In order to reduce the number of roots we can impose a constraint like  $z_+ z_- = \Phi \neq 0$ , for some number complex number  $\Phi$ .

**Theorem 2.** *The basic unipodal equation  $z^n = r$ , subject to the constraint  $z_+ z_- = \Phi$ , has the  $n$  solutions*

$$z = (e^{\frac{i2\pi}{n}})^k r_+^{\frac{1}{n}} u_+ + \frac{\Phi u_-}{(e^{\frac{i2\pi}{n}})^k r_+^{\frac{1}{n}}}$$

where  $k = 0, 1, 2, \dots, n-1$  and  $r_{\pm}^{\frac{1}{n}}$  is any  $n^{\text{th}}$  root of  $r_{\pm}$ .

Proof If we let  $z_+ z_- = \Phi$  that is the same as  $z_- = \frac{\Phi}{z_+}$ . applying this above we obtain the result.

With this in hand we can state the solution to any depressed cubic.

**Theorem 3.** *The depressed cubic  $x^3 + 3ax + b$ , for  $k = 0, 1, 2$ , has roots*

$$x = \frac{1}{2} \left( \omega^k \sqrt[3]{-4b + 4\sqrt{b^2 + 4a^3}} + \frac{-4a}{\omega^k \sqrt[3]{-4b + 4\sqrt{b^2 + 4a^3}}} \right)$$

Proof Starting with the unipodal equation  $z^3 = r = s + tj$ , it can be written in the standard basis as

$$\begin{aligned} (x + jy)^3 &= s + tj \\ \iff (x^3 + 3xy^2) + (3x^2y + y^3)j &= s + tj. \end{aligned}$$

Equating scalar parts we get

$$x^3 + 3xy^2 - s = 0.$$

In order eliminate  $y$  we impose the condition that  $z_- z_+ = \Phi \iff x^2 - y^2 = \Phi$

$$\iff y^2 = x^2 - \Phi \iff x^3 + 3x(x^2 - \Phi) + s = 0$$

$$\iff x^3 - \frac{3\Phi}{4}x - \frac{s}{4} = 0,$$

If we go back to our constraint we also get

$$\Phi^3 = (z_+ z_-)^3 = z_+^3 z_-^3 = r_+ r_- = s^2 - t^2$$

so  $t = \sqrt{s^2 - \Phi^3}$ . We want our equation  $x^3 - \frac{3\Phi}{4}x - \frac{s}{4}$  to be equal to the depressed cubic, so we let  $\Phi = -4a$  and  $s = -4b$ , which gives  $t = 4\sqrt{b^2 + 4a^3}$ .

Thus the scalar part of the solution to  $z^3 = -4b + 4j\sqrt{b^2 + 4a^3}$  with our choice of  $\Phi$  solves the depressed cubic. We know from Theorem 2 that  $z$  is

$$\begin{aligned} z &= \omega^k (-4b + 4\sqrt{b^2 + 4a^3})^{\frac{1}{3}} u_+ + \frac{-4a}{\omega^k (-4b + 4\sqrt{b^2 + 4a^3})^{\frac{1}{3}}} u_- \\ &= \frac{1}{2} (\omega^k (-4b + 4\sqrt{b^2 + 4a^3})^{\frac{1}{3}} (1 + j) + \frac{-4a}{\omega^k (-4b + 4\sqrt{b^2 + 4a^3})^{\frac{1}{3}}} (1 - j)) \\ &= \frac{1}{2} \left[ \omega^k (-4b + 4\sqrt{b^2 + 4a^3})^{\frac{1}{3}} + \frac{-4a}{\omega^k (-4b + 4\sqrt{b^2 + 4a^3})^{\frac{1}{3}}} \right] + \frac{j}{2} \left[ (4b + 4\sqrt{b^2 + 4a^3})^{\frac{1}{3}} - \frac{-4a}{\omega^k (-4b + 4\sqrt{b^2 + 4a^3})^{\frac{1}{3}}} \right] \end{aligned}$$

for  $k = 0, 1, 2$ , which delivers us the result.

Returning back to our original problem when we obtained  $x^3 + \frac{8}{3}x - \frac{2}{27}$ , we have  $a = \frac{8}{9}$  and  $b = -\frac{2}{27}$ , so

$$x = \frac{1}{2} \left[ \omega^k \left( \frac{8}{27} + 4\sqrt{\frac{4}{729} + \frac{2048}{729}} \right)^{\frac{1}{3}} + \frac{\frac{-32}{9}}{\omega^k \left( \frac{8}{27} + 4\sqrt{\frac{4}{729} + \frac{2048}{729}} \right)^{\frac{1}{3}}} \right]$$

Simplifying this arithmetically we get

$$\begin{aligned} x &= \frac{1}{2} \left[ \omega^k \sqrt[3]{\frac{8 + 24\sqrt{57}}{27}} - \frac{\frac{-32}{9}}{\omega^k \sqrt[3]{\frac{8 + 24\sqrt{57}}{27}}} \right] \\ &= \frac{1}{3} \left[ \omega^k \sqrt[3]{1 + 3\sqrt{57}} - \frac{8}{\omega^k \sqrt[3]{1 + 3\sqrt{57}}} \right] \end{aligned}$$

which for  $k = 0, 1, 2$  numerically evaluate to 1 real root 0.2777, and two complex roots  $-0.0139 + 1.6332i$  and  $-0.0139 - 1.6332i$ .

We can confirm that these are the same answers as the first method not just numerically, but also by

$$\begin{aligned} \frac{1}{3} \left[ \sqrt[3]{1 + 3\sqrt{57}} - \frac{8}{\sqrt[3]{1 + 3\sqrt{57}}} \cdot \frac{\sqrt[3]{1 - 3\sqrt{57}}}{\sqrt[3]{1 - 3\sqrt{57}}} \right] &= \frac{-\sqrt[3]{-1 - 3\sqrt{57}} - \sqrt[3]{-1 + 3\sqrt{57}}}{3}, \\ \frac{1}{3} \left[ \omega \sqrt[3]{1 + 3\sqrt{57}} - \frac{8}{\omega \sqrt[3]{1 + 3\sqrt{57}}} \cdot \frac{\omega^2 \sqrt[3]{1 - 3\sqrt{57}}}{\omega^2 \sqrt[3]{1 - 3\sqrt{57}}} \right] &= \frac{-\omega \sqrt[3]{-1 - 3\sqrt{57}} - \omega^2 \sqrt[3]{-1 + 3\sqrt{57}}}{3}, \\ \frac{1}{3} \left[ \omega^2 \sqrt[3]{1 + 3\sqrt{57}} - \frac{8}{\omega^2 \sqrt[3]{1 + 3\sqrt{57}}} \cdot \frac{\omega \sqrt[3]{1 - 3\sqrt{57}}}{\omega \sqrt[3]{1 - 3\sqrt{57}}} \right] &= \frac{-\omega^2 \sqrt[3]{-1 - 3\sqrt{57}} - \omega \sqrt[3]{-1 + 3\sqrt{57}}}{3}. \end{aligned}$$